TECHNICAL DATASHEET

Alpha Staircase Tower

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AST-4 • 4-Leg Cuplock

Square Pipe Grating

Prime Steeltech (I) Pvt. Ltd.
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AST-4 • 4-Leg Cuplock | M.S. Square Pipe Grating | ISO 9001 Certified | Self-Draining

Cuplock scaffolding engineered for self-draining performance.

The Alpha Staircase Tower AST-4 is engineered for sites where drainage is critical — basement excavations, below-grade construction, open-air infrastructure, and wet or muddy environments.

The AST-4 configuration (4-Leg Cuplock) features M.S. Square Pipe Grating treads and M.S. Rectangular Pipe Grating landing platforms that eliminate mud and water accumulation for safer access in adverse conditions.

The 60 mm OD welded socket joint provides positive vertical continuity without loose spigot pins, accelerating erection and improving structural integrity.

Full fall protection is provided on all stair flights via side railings at 940 mm and mid-rails at 470 mm, with the uppermost platform protected by railings at 1,100 mm and mid-rails at 750 mm. Adjustable base jacks with 450 mm travel and mandatory tie bars ensure secure installation on any terrain.

AST-4

Pipe Steel Grade	IS 1161 Grade YST210
Tread Surface	M.S. Square Pipe Grating — 25 × 25 mm
Landing Surface	M.S. Rectangular Pipe Grating — 25 mm
Paint Finish	Red-Oxide Primer + Blue Enamel (Standard)
Grid Size	1800 x 2500 mm

Overall Tower Geometry

GRID SIZE (L × W) 2,500 × 1,800 mm	**PLATFORM HEIGHT** Up to 60,000 mm (60.0 m)
TOWER OVERALL HEIGHT Platform height + 1,500 mm (top rail)	**TOP RAILING HEIGHT** 1,500 mm above platform
NO. OF VERTICAL MEMBERS 4	**CONFIGURATION** AST-4, 4-leg cuplock

Stair Geometry

Parameter	Specification
Stair Width (clear)	700 mm
Stair Type	Zig-Zag Step Arm
Stair Material	M.S. Square Pipe Grating — 25 × 25 mm (open mesh, self-draining)
Side Railing Height (Stair Flights)	940 mm — both sides, M.S. Square Pipe
Guard Rail / Mid Rail (Stair Flights)	470 mm — both sides, M.S. Square Pipe
Top Platform: Side Railing Height	1,500 mm — both sides
Top Platform: Guard Rail (Mid Rail)	1000 mm — both sides
Step Rise	237 mm (Class-B per EN 12811-1; BS 1139-3)
Tread (going)	270 mm (Class-B per EN 12811-1; BS 1139-3)
Staircase Angle	49° (EN 12811-1: 30-55°; OSHA 1926.451: 40-60°)
Step Arm UDL (Design Load)	250 kg/m² (2.45 kN/m² | 50.1 psf)
Side rails Both side	Yes
No. of stairs per flight	5

Landing Platform — Quick Reference

LANDING WIDTH 1450 mm	**LANDING DEPTH** 1000 mm
TOE GUARDS Not provided — economy configuration	**LANDING FIXING** Hook-on, secure
PLATFORM UDL 250 kg/m²	**SURFACE** M.S. Rectangular Pipe Grating

Structural Framing

Component	Specification
Cuplock Vertical (Standard)	48.3 OD × M.S. IS 1161 Grade YST210
Top Cup	SG / Malleable Cast Iron
Bottom Cup	Pressed Metal
Cup Spacing	500 mm centers
Cuplock Ledger	48.3 OD M.S. pipe, forged ledger blade each end
Vertical Joint (Socket)	60 mm OD socket, welded onto vertical standard
Bracing Pipe	48.3 OD M.S. pipe, all 4 sides
Couplers & Clamps	Heavy-duty forged; IS 2750 / EN-74 compliant

Base & Anchorage

Component	Specification
Adjustable Base Jack	Spindle: 38 mm OD; Total: 600 mm; Adjustable: 450 mm; Base Plate: 150 × 150 × 5 mm M.S.
Tie Bar	48 mm OD M.S. pipe + welded holding plate; anchor fastening provision 150 x 150 x 6 mm ; secures tower to RCC structure / walls

All dimensions in mm unless noted. Weights subject to ±3.0% tolerance. Structural design required for platform heights exceeding 10 m.

Applicable Standards & Design Codes

Standard	Scope / Relevance
IS 1161:2014	Steel tubes for structural purposes — OD, wall thickness & mechanical properties (YST210/240/310) for all M.S. tubes.
IS 2750:1964	Steel scaffolding — fittings. Drop-forged and pressed couplers and clamps.
IS 4014 (Pt.1):1967	Steel tubular scaffolding — materials. Component dimensions, grades, workmanship.
IS 4014 (Pt.2):1967	Steel tubular scaffolding — safety regs. Stairway access, handrails, tie-bar spacing, bracing.
IS 3696 (Pt. 1): 1987	Safety code for scaffolds & ladders — stairway design, railings, toe-boards, platform protection
ISO 9001:2015	Quality management systems — Prime Steeltech India QMS certification.

Key Features & Applications

SAFETY NOTICE — TOE BOARDS NOT PROVIDED (ECONOMY CONFIGURATION). Toe boards on intermediate landings are NOT provided in this economy configuration. The contractor and site engineer must ensure supplementary measures to prevent tools and materials falling from landing edges are in place before permitting personnel access. A project-specific risk assessment by a competent person is mandatory.

Cuplock System

Tool-free assembly — cup locks up to 4 ledgers per node simultaneously.

Rectangular Pipe Grating Platforms

M.S. 25 mm open-mesh landings; lightweight, low self-weight, easy site handling.

Self-Draining Construction

Open grating ideal for basement, excavation, and exposed wet-site applications.

Adjustable Base Jack

150 × 150 × 6 mm plate; 450 mm adjustable travel; levels on uneven ground.

Platform UDL 250 kg/m²

Design load 2.45 kN/m² (50.1 psf); suited for personnel and tools access.

Square Pipe Grating Treads

M.S. 25 × 25 mm open-mesh; self-draining — eliminates mud and water accumulation on treads.

Welded 60 mm Socket Joint

Positive vertical continuity; faster erection with no loose spigot pins.

Full Fall Protection (Stair Flights)

Side railing @ 940 mm & mid-rail @ 470 mm in M.S. square pipe; both sides of all stair flights.

Tie Bar to RCC / Walls

48 mm OD tie bar with anchor fastening provision; mandatory for height > 6.0 m.

Typical Applications

- Basement excavation & below-grade construction
- High-rise residential & commercial construction
- Infrastructure: bridges, flyovers, metro viaducts
- Demolition & decommissioning works
- Aircraft maintenance hangar scaffolding
- Industrial plants: refineries, petrochemical, power, cement
- Open-air & exposed infrastructure — drainage critical
- Oil & gas platforms, jetty structures
- Shipyards & dockyard construction
- Short-duration construction access & temporary platforms

Safety Guidelines

SAFETY NOTICE — Never exceed the Safe Working Load (SWL) for the grating and span combination. Guardrails min 1,000 mm height must be installed before access at height. Scaffold erection, alteration and dismantling by trained personnel only (IS 3696). Conduct a project-specific risk assessment before permitting personnel access.

1. Inspect every grating panel before installation — remove cracked, bent or heavily corroded units immediately.
2. Ensure hooks / end fittings on landing platforms are fully seated before personnel access.
3. Toe boards are NOT provided in this economy configuration — supplementary measures must be in place before permitting access to landing edges.
4. Do not mix grating types / sizes within the same bay without written engineering approval.
5. M.S. grating surface must be free of mud, oil, ice or loose debris at all times.
6. Display inspection tags (load class, date, inspector name) at each access point.
7. Field welding of grating panels is NOT permitted — return damaged panels for factory repair.
8. Tie bars are mandatory once height exceeds 6.0 m; install at maximum 3.0 m vertical and horizontal spacing. Spacing must be confirmed by a structural engineer.

Notes & Disclaimers

1. Staircase towers must be designed, erected, and inspected by competent persons in accordance with IS 3696 Part 1, IS 4014 Parts 1 & 2, and the BOCW Act 1996.
2. Toe boards on intermediate landings are NOT provided in this economy configuration. Supplementary measures to prevent falling objects must be in place before permitting access.
3. Adjustable base jack spindle: 38 mm OD pipe; total length 600 mm; usable / adjustable length 450 mm. Must not be extended beyond usable length; use sole boards on soft or unpaved ground.
4. M.S. grating treads and platforms must be inspected regularly for corrosion, deformation, weld cracking, or mechanical damage. Damaged panels must be replaced before further access.
5. Platform and step-arm UDL of 250 kg/m² (2.45 kN/m²) is the maximum design value. Site loadings must not exceed this value.
6. This datasheet is for information and quotation reference only. Structural calculations and certified drawings are prepared separately. Contact Prime Steeltech India for project-specific design.

Contact & Further Information

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